

• n -pt vertex : $V_n(k_1, \dots, k_n)$

* For ϕ^3 :

(i) $V_{n>3}$: there is no tree contribution to $V_{n>3}$

(ii) $V_{n>3}$: One-loop contributions to $V_{n>3}$ is finite for $d < 2n$

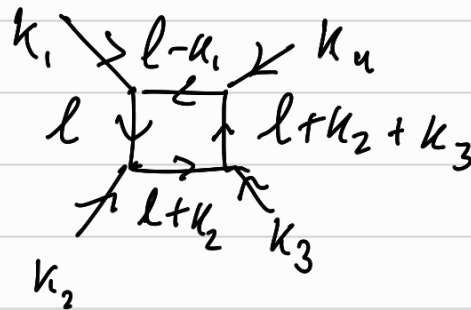
* Let $n=4$:

• All external momenta is INCOMING:

$$k_1 + k_2 + k_3 + k_4 = 0$$

(S-channel)

$$iV_4^S(k_1, \dots, k_4) \sim$$



$$iV_4^S(k_1, \dots, k_4) = (ig)^4 \left(\frac{1}{i^4}\right) \int \frac{d^d l}{(2\pi)^d} \frac{\tilde{\Delta}(l^2) \tilde{\Delta}(l+k_2) \tilde{\Delta}(l+k_2+k_3)}{\tilde{\Delta}(l-k_1)^2}$$

$$\tilde{\Delta}\tilde{\Delta}\tilde{\Delta}\tilde{\Delta} = \int dF_4 (x_1 l^2 + x_2 (l+k_2)^2 + x_3 (l+k_2+k_3)^2 + x_4 (l-k_1)^2 + m^2)^{-4}$$

$$= \int dF_4 (x_1 l^2 + x_2 (l^2 + 2l \cdot k_2 + k_2^2) + x_3 (l^2 + 2l \cdot (k_2+k_3) + (k_2+k_3)^2) + x_4 (l^2 - 2l \cdot k_1 + k_1^2) + m^2)^{-4}$$

$$= \int dF_4 [l^2 + 2l \cdot (x_2 k_2 + (k_2+k_3)x_3 - x_4 k_1) + x_2 k_2^2 + x_3 (k_2+k_3)^2 + x_4 k_1^2 + m^2]^{-4}$$

$$= \int dF_4 [(\ell + x_2 k_2 + x_3 (k_2+k_3) - x_4 k_1)^2 + (x_2 k_2 + x_3 (k_2+k_3) - x_4 k_1)^2 + x_2 k_2^2 + x_3 (k_2+k_3)^2 + x_4 k_1^2 + m^2]^{-4}$$

$$Q \equiv \ell + x_2 k_2 + x_3 (k_2+k_3) - x_4 k_1$$

$$= \int dF [Q^2 + D]^{-4}$$

$$D_{1234} = x_2 k_2^2 + x_3 (k_2 + k_3)^2 + x_4 k_1^2 + x_2^2 k_2^2 + x_2 x_3 k_2 (k_2 + k_3) - x_2 x_4 k_2 k_1$$

$$= x_1 x_4 k_1^2 + x_2 x_4 k_2^2 + x_2 x_3 k_3^2 + x_1 x_3 k_4^2 + x_1 x_2 (k_1 + k_2)^2 + x_3 x_4 (k_2 + k_3)^2 + k_1^2$$

using $\sum_i x_i = 1$ and $\sum_i k_i = 0$

* Then

$$iV_4^S = g^4 \int dF_n \int \frac{d^d q}{(2\pi)^d} \frac{1}{(q^2 + D)^4} + \mathcal{O}(g^6)$$

Wick rotate $q_0 \rightarrow i q_d$

$$iV_4^S = i g^4 \int dF_n \int \frac{d^d \vec{q}}{(2\pi)^d} \frac{1}{(\vec{q}^2 + D_{1234})^4}$$

Converges if $d - 8 < 0$
 $d < 8$

and so $d=6$, where $[g]=0$, is applicable without needing to regularize the mass

dim:

$$\int \frac{d^6 \vec{q}}{(2\pi)^6} \frac{1}{(\vec{q}^2 + D)^4} = \frac{\Gamma(4 - 6/2)}{(4\pi)^{d/2} \Gamma(4)} D^{-(4 - 6/2)}$$

$$= \frac{\Gamma(1) D^{-1}}{(4\pi)^3 \Gamma(4)} = \frac{D_{1234}}{6(4\pi)^3}$$

$$V_4 = V_4^S + V_4^t + V_4^u = \frac{g^4}{6(4\pi)^3} \int dF_4 \left(\frac{1}{D_{1234}} + \frac{1}{D_{1324}} + \frac{1}{D_{1243}} \right) + \mathcal{O}(g^6)$$

* Which is finite and well-defined